Freshness Forecast

Snap your produce. Know how many days it has left.



AI/ML Project Proposal | Geek Room Society

The Problem & Who Faces It

The problem

Fruit and vegetables spoil fast, and it is hard to judge by eye how much time is left. People either throw away food that was still fine, or eat it too late. Existing apps only say “fresh” or “rotten”, not how many days remain. Without that, nobody can plan meals, shop smartly or store produce well.

Who faces it

- **Households**
  Families buying fresh groceries weekly
- **Students & young professionals**
  Hostels, PGs and busy schedules
- **Fruit & vegetable vendors**
  Small shops deciding what to discount
- **Canteens & cloud kitchens**
  Bulk produce on tight margins

Why This Is Worth Solving

55 kg

of food wasted per person each year by Indian households

60%

of global food waste happens at household level

8–10%

of global greenhouse-gas emissions come from food loss and waste

If it isn’t solved

Money
Groceries bought, then binned

Environment
Food in landfills releases methane

Health
Spoiled produce eaten by mistake

Our Solution: A Freshness Forecast

1. Snap And Tell

Photo of fruit or veg with a phone camera And Tap where it is kept: Fridge or Counter.

2. Recognise

AI finds the produce type and decay stage

3. Forecast

Predicts how many days it stays good

4. Act

Use-first alert, storage tip, recipe idea

EXAMPLE OUTPUT (ILLUSTRATIVE)

Apple And in fridge | Ripe | about 5 days left.| Keep it in the fridge to stretch it Or you want any reciepe .

Where AI/ML Comes In

1. Identify the produce

Image classification: which fruit or vegetable?

2. Judge the decay stage

Classification: fresh, ripe, overripe, spoiled

3. Forecast days left

The model takes the photo details, where the item is kept (fridge or counter) and **today's weather**. From these it predicts how many days the item will stay good.

ML types involved

- Computer Vision
- Classification
- Regression

What Data We Need

1. Images

Fresh vs rotten fruit photos (Kaggle, Mendeley) to start

2. Time-lapse photos

Our own daily photos of one item until it spoils

3. Storage conditions

Fridge or counter, temperature, humidity (tabular)

4. Produce reference

Known shelf-life tables for each fruit and vegetable

5. Location & time

Season and local weather, since heat speeds spoilage

6. User feedback

“Was the forecast right?” taps to improve the model

What Exists & What's Different

Similar work

BananaClock
Open-source web app: banana ripeness stage and a days-until-bad countdown. Bananas only.

Hass avocado study (2024)
CNN predicts ripeness and shelf life to within about a day. One fruit, lab setup.

WUR ShelfLifeCalculator
Predicts shelf life from storage conditions. Built for growers and traders.

Our edge

1.One app for many foods. Others work for only one fruit. Ours checks fruits and vegetables of all kinds.

2.Photo plus storage. We also ask where the item is kept (fridge or counter), so the forecast is more accurate.

3.Tells you what to do. We don't just say "fresh" or "rotten." We say "eat first," "store it," or "cook it."

4.Made for everyday people. It is for homes and small vendors, and it runs on a budget phone.

Challenges & Open Questions

1. Data availability
No public set has days-left labels; we must photograph and log it

2. Accuracy
Lighting, backgrounds and local varieties can confuse the model

3. Hidden spoilage
Rot inside the fruit isn't visible, and storage changes the pace

4. Privacy
Photos can stay on the phone; keep personal data to a minimum

5. Compute
Must run fast on budget phones, so small networks only

6. Deployment
Putting the app online and retraining the model with user feedback must be done without breaking the app.

Sources & Research

Flutter app for fruit shelf-life prediction (mango)
https://ideas.repec.org/a/igg/jirr00/v12y2022i3p1-20.html

Kaggle: Fruits fresh and rotten for classification
https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification

Kaggle: Fresh and stale images of fruits and vegetables
https://www.kaggle.com/datasets/raghavrpotdar/fresh-and-stale-images-of-fruits-and-vegetables

BananaClock: banana ripeness and shelf-life web app
https://peerlist.io/dsharma08k/project/bananaclock